EXAMPLE UNIVERSITY - MID-SEMESTER EXAMINATION 2024
School of Engineering (synthetic sample paper)
Subject: Sample Systems Modelling (XX3001) (Regular)
> 4 (four) questions are to be attempted.
> Section A is compulsory; attempt any 1 question from each of Sections B, C and D.
Time: 1.5 hours Full Marks: 20
The figures in the right-hand side indicate full marks.

State the main stages of a closed-loop feedback controller.

Why does gradient descent become unstable when the step size is too large?

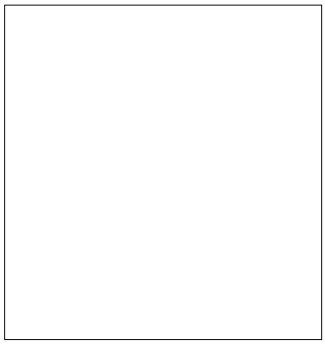
Name the smallest network depth that can separate the XOR pattern.

Why are smooth squashing functions chosen as the activation in a layered model?

A set is given as A={0.2/1, 0.4/2, 0.5/4, 0.9/5, 0.3/6, 0.5/7}. State whether it is convex. Justify.

Consider a universe X of ages X = {5, 15, 25, 35, 45, 55, 65, 75, 85}.

Answer the following
1. Build the sets named young, adult and senior over X.
2. Give the support of each set.
3. What is the width of the set young?
4. Is the set adult normal or subnormal?
5. What is the core of the set senior?



Let M be a set describing how mature a person is at a given age. Its membership rule is
$\mu_M(x) = \begin{cases}0, if\ x \leq 5\\\left(\frac{x-5}{20}\right)^2, if\ 5 < x \leq 25\\1, if\ x > 25\end{cases}$
The universe is X={3, 8, 10, 12, 15, 18, 20, 24, 27, 32}. Find the alpha-cut and the strong alpha-cut for alpha=0.4.

Sketch the block model of a single artificial unit and mark every unknown parameter.

A two-class training set is listed below. Find the weights after the first iteration, starting from W=[1 -0.5].



Write down the learning rule of a linear adaptive unit.

With starting weights W=[0.2, 0.3], update the weights for the input X=[-1 -1] when the target is d=-1.

How does a radial basis layer turn a nonlinearly separable task into a linearly separable one?

Use the XOR function with Gaussian basis functions to illustrate your answer.

What is meant by the cooperation step in a self organising map?

A map has two inputs and two output units with initial weights
$W = \begin{bmatrix}0.2 & 0.3\\0.7 & 0.4\end{bmatrix}$
Find the output unit closest to the input [0.4 0.3] and update its weights with a learning rate of 0.5.